

# Native SSH and SFTP Support in Unicon

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## Abstract

Unicon historically had no native SSH client: programs that needed remote command execution or file transfer shelled out to an external ssh binary. This report describes the current design and implementation: a client that binds libssh and exposes sessions, multiplexed channels, and SFTP through existing language verbs – `open()`, `read()/write()`, `receive()`, `Attrib()`, `stat()`, `remove()`, and `rename()` – rather than a parallel SSH-specific API. The report covers the connection and channel model, authentication and attributes, examples and use cases, the three I/O tiers, interactive shells, SFTP, build integration, and remaining work.

**Keywords:** Unicon, SSH, SFTP, libssh, sockets, runtime, technical report.

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# 1 1. Introduction

This report describes the current design and implementation of native SSH in the Unicon runtime: how a session is opened, how additional channels and SFTP handles are derived from it, how stream I/O, ordered events, and metadata operations are exposed, and what remains out of scope. It is formatted as a Unicon Technical Report [1]. The language-facing reference lives in *Programming with Unicon* (Chapter 6, "Secure Shell") and the language reference (`open` mode `h`, `Attrib`, `receive`, `stat` / `remove` / `rename`). section 6 collects examples by mode and use case. Automated tests live in `tests/posix/ssh.icn`.

The feature is optional. A build with `libssh` present reports `secure shell` in `&features` (the `_SSH` feature flag). Without the library, mode `h` is ignored and the SSH paths are compiled out, as with other optional features.

## 2 2. Motivation and prior art

Unicon programs that needed SSH previously relied on `popen()` or pipes to an external `ssh` process. That is not portable (Windows often has no OpenSSH client in `PATH`) and gives the program no structured access to channels, exit status, or SFTP. Native primitives let an application authenticate to hosts, run commands, and transfer files without an external process.

### 2.1 2.1 Library choice

Two C libraries were considered. `libssh` (LGPL) supports both client and server, modern key exchange and host keys, and true non-blocking I/O via `poll(2)`. `libssh2` (BSD) is client-only and cannot be used fully non-blocking. The implementation binds `libssh` [2], for server-side optionality later and for the callback API that preserves `stdout/stderr` arrival order ([section 8.3](#)).

A full protocol reimplementaion (Paramiko, Go's `x/crypto/ssh`) would mean owning crypto primitives and protocol-drift / CVE tracking indefinitely. Binding to `libssh` via the existing C runtime avoids that.

## 2.2 2.2 Open as the verb

Python (Paramiko) and Go (`x/crypto/ssh`) both treat "new session" as sugar over "open another channel on this connection." The verb is **open**, applied to an existing connection object, not "clone" or "duplicate." Unicon already has `open()` as the universal constructor for files, sockets, SSL, and messaging connections. `Clone()` is the Graphics-facility naming convention; a distinct `channel()` function would add a new global name for an operation `open()` already expresses.

## 2.3 2.3 Why not Messaging

The Messaging facility (HTTP/POP/SMTP, `rmsg.r`, `libtp`) [3] is not a fit. Messaging's transport-discipline abstraction has the same 1:1 single-stream limitation as SSL – no channel-multiplexing concept – so moving into that world would not solve the multiplexed-channel requirement, only relocate it. SSH's request/channel model (`exec` / `shell` / `sftp` as concurrent channels) also does not map onto Messaging's request/response verb model (`GET` / `POST` / `RETR`). SSH does not route through `ssh://` URIs or `Mopen()`. It uses the `h` mode character and plain `user@host` string parsing, independent of the Messaging/URI subsystem.

## 3 3. Architectural grounding

The implementation follows established SSL and socket patterns rather than inventing a parallel subsystem.

`union f` in `src/h/rstructs.h` is the file block's descriptor slot: a tagged union of `FILE *`, socket `fd`, `SSL *` when `HAVE_LIBSSL`, and so on, discriminated by status bits at runtime. SSH does **not** embed `libssh` objects directly in the union. It adds a pointer, `struct SSHfile *sshf`, matching the Messaging `MFile *` precedent. `SSHfile` is allocated with `malloc` so pointers to it are stable across garbage collection, and is freed by the close hook, never by the collector. A `malloc`'d block is required once the session must hold a list of child channels whose addresses survive compaction ([section 10.2](#)).

Status bits live in `src/h/rmacros.h`. `Fs_Encrypt` is 0200000000; SSH takes the next bit, `Fs_SSH` = 0400000000. `open()`'s mode string is scanned character-by-character in `src/runtime/fsys.r`. `e` sets `Fs_Encrypt`. SSH uses `h` / `H` (`s` is already the Messaging short-request flag).

Attributes are "key=value" strings passed as trailing arguments to `open()`, parsed in `create_ssh_session()` the same way `create_ssl_context()` parses SSL attributes – not a fixed argument list. A leading integer, when present, is a connect timeout in milliseconds, as for plain sockets.

All stream I/O funnels through one dispatch point: `u_read()` / `u_write()` in `src/runtime/rposix.r`, which already branch on `Fs_Socket` / `Fs_Encrypt`. SSH adds a third branch for `Fs_SSH`, reading from the arrival-order queue (section 8) or `sftp_read()`, and writing via `ssh_channel_write()` or `sftp_write()`. Nothing above this dispatch (`reads()`, `read()`, and so on) needs its own SSH case, except where a line-oriented helper must not wait for a newline that will never arrive (section 9.2).

Close is **explicit, not GC-driven**. The close hook in `fsys.r` frees SSL resources directly; there is no finalizer during garbage collection. SSH files follow the same convention – leaking a channel or session means not calling `close()`, the same as any other file.

`receive()` already returns a general-purpose `posix_message` record (fields `addr` and `msg`) for UDP and raw datagrams. SSH reuses that type for an ordered stdout/stderr/exit-status event stream (section 8.3).

`stat(f)` already `type_cases` on its argument (string path versus open file). SFTP extends both branches (section 11.3). `remove()` and `rename()` gain an optional leading session argument, matching the existing Unicon idiom that many functions take an optional leading file or window (`write("hello")` versus `write(f, "hello")`).

## 4 4. Connection and channel model

### 4.1 4.1 open() dispatch

Two shapes, both using the existing `open()`. The second argument is the usual mode string; `h` is a new character in that scan, and `c` / `s` after `h` select a channel or SFTP the same way `u` after `n` selects UDP (`n` is TCP, `nu` is UDP).

- `open(host_or_"user@host", "h...", attrs...)` – first argument is a string: opens a new authenticated SSH **session** and, by default, an interactive shell channel on it (section 4.3).

- `open(s, "hc"/"hs", attrs...)` – first argument is an existing SSH session or channel file: opens a **new channel** (`hc`) or SFTP handle (`hs`) on the same underlying session (reusing the transport and authentication, no new handshake). Opening on a channel opens a **sibling** on the session owner, not a child of the channel.

A session opened with `channel=no` has no channel of its own (`Fs_SSH` alone, no `Fs_Socket`). It exists so later `open(s, "hc")` / `open(s, "hs")` calls have an authenticated transport. `channel=yes` is the default. Combining `channel=no` with mode `hc` or `cmd=` is contradictory and fails with a bad-attribute error.

```
s := open("jafar@" || host, "h", "key=" || keypath)
write(s, "show version\n")
while line := reads(s) do write(line)
close(s)

s := open("user@host", "h", "key=id_ed25519", "channel=no")
c1 := open(s, "hc", "cmd=uname -a")
c2 := open(s, "hc", "cmd=uptime")
```

## 4.2 4.2 Host string

The first argument is `host`, `user@host`, or either with `:port`. The port uses the same `host:port` form as socket `open()`; IPv6 uses `[addr]:port`. A bare IPv6 literal is not misparsed: an unbracketed host is split on `:` only when there is a single colon. There is no `port=` attribute.

`user@host` wins over a `user=` attribute. If neither is given, libssh's default user (the local login name) is used.

`ssh://user@host` URI syntax is not used ([section 2.3](#)).

## 4.3 4.3 Default channel: shell, not bare session

`open(host, "h", ...)` with no `cmd=` and without `channel=no` opens a shell on a remote PTY immediately, so the common interactive case is one handle. The PTY is requested with `TERM` (from `term=`, else `$TERM`, else `xterm`) and a window size (from `cols=` / `rows=`, else the local tty via `TIOCGWINSZ`, else 80x24). Without a real size, remote `ls(1)` and friends fall back to one column.

A shell channel has no clean command boundaries: prompt text, echoed input, and output all interleave. For scripted use, prefer mode `hc` with `cmd=`, which has clean stdout and a real exit status:

```
c := open(s, "hc", "cmd=show version")
```

## 5 5. Attributes and authentication

Trailing `open()` arguments are `name=value` strings. Empty values and unknown names fail (error 1321). Boolean-style attributes without a value are not accepted – every attribute must contain `=`. SFTP is selected by mode character `s` after `h` ("`hs`"), not by a valueless attribute.

| Attribute                  | Meaning                                          | Notes                                                             |
|----------------------------|--------------------------------------------------|-------------------------------------------------------------------|
| <code>user=</code>         | Remote login name                                | Overridden by <code>user@host</code>                              |
| <code>key=</code>          | Path to a private key file                       | Same convention as SSL's <code>key=</code>                        |
| <code>keypass=</code>      | Passphrase for an encrypted private key          | Same name as TLS/crypto; distinct from                            |
| <code>password=</code>     | Remote login password                            | Not a TLS attribute; crypto still accepts                         |
| <code>verifyPeer=</code>   | <code>yes</code> (default) or <code>no</code>    | Same spelling as TLS; overrides mode                              |
| <code>hostkeyfile=</code>  | Path to an OpenSSH <code>known_hosts</code> file | Passed to libssh as <code>SSH_OPTIONS_KNOWN_HOSTS</code>          |
| <code>cmd=</code>          | Command for a one-shot exec channel              | Requires mode <code>hc</code> ; contradicts <code>channel=</code> |
| <code>channel=</code>      | <code>yes</code> (default) or <code>no</code>    | <code>no</code> is transport-only                                 |
| <code>term=</code>         | Remote <code>TERM</code> for a PTY               | Default <code>\$TERM</code> or <code>xterm</code>                 |
| <code>cols= / rows=</code> | PTY size                                         | Default from the local tty, else 80x24                            |

`keypass=` is the shared name for a key-file passphrase on TLS, crypto, and SSH. SSH `password=` remains the remote login password and is not accepted on TLS. Crypto still accepts `password=` as an alias for `keypass=`.

The `-` mode-character flag disables peer verification on TLS (`ne-`) and SSH (`h-`), the same as `verifyPeer=no`. An explicit `verifyPeer=` overrides the mode flag. `Fs_Encrypt` and `Fs_SSH` are mutually exclusive contexts, so there is no conflict.

A leading integer attribute is a connect timeout in milliseconds and is applied as libssh's `SSH_OPTIONS_TIMEOUT / TIMEOUT_USEC`. The default port is 22 when `:port` is omitted.

## 5.1 5.1 Authentication order

If both `key=` and `password=` are given, they are tried in the order the attributes appear in the call, not in a hardcoded priority. If **neither** is given, the implementation falls back to `ssh_userauth_publickey_auto()` – default identities under `~/.ssh` and a running agent – not to password authentication. Agent use through this fallback is supported; an explicit `agent=` attribute and keyboard-interactive auth remain out of scope.

`key=` does not expand `~`. Callers (and the `ussh` demo) must expand a leading `~/` themselves; `libssh` treats the path literally.

## 6 6. Examples and use cases

These examples assume a Unicon build with the `secure shell` feature. Connection failures fail with `&errortext` set; the usual form is `s := open(...) | stop(&errortext)`. The packaged demo `uni/progs/ussh.icn` is a small client that covers the interactive and one-shot command cases.

### 6.1 6.1 Modes

Like every other `open()`, SSH uses the second argument as a mode string (it defaults to `"r"` when omitted, which is not useful for SSH). `h` / `H` is a mode character in that same scan, alongside `r` / `w` / `n` / `e` / `m`. After `h`, `c` selects a command/channel and `s` selects SFTP – the same "facility letter, then modifier" order as `nu` (UDP) and `ms` (Messaging short-request). `n` alone is TCP. Without a preceding `h`, `c` is still `create+write` and `s` is still ignored (or Messaging short-request after `m`). Trailing `"name=value"` arguments are attributes, not modes – the same slot SSL uses for `key=` and friends.

Mode characters combine. `"h-"` is SSH plus skip host-key verification (the `-` flag also disables SSL peer checks). `"hc"` is a channel (exec when `cmd=` is given, else a shell). `"hs"` is SFTP; `r` / `w` / `a` / `b` set transfer intent (`"hsw"`, or `open(..., "hs", ..., "w")`).

| Form                                       | Kind                                                  | Handle               |
|--------------------------------------------|-------------------------------------------------------|----------------------|
| <code>open(host, "h", ...)</code>          | mode <code>h</code>                                   | New session plus a P |
| <code>open(host, "h-", ...)</code>         | mode <code>h</code> and <code>-</code>                | Same, no host-key cl |
| <code>open(host, "hc", "cmd=...")</code>   | mode <code>hc</code> , attribute <code>cmd=</code>    | New session plus an  |
| <code>open(host, "h", "channel=no")</code> | mode <code>h</code> , attribute <code>channel=</code> | Session only (no cha |



| Form                                           | Kind                                               | Handle                |
|------------------------------------------------|----------------------------------------------------|-----------------------|
| <code>open(s, "hc")</code>                     | mode <code>hc</code> on an existing SSH file       | Extra shell channel o |
| <code>open(s, "hc", "cmd=...")</code>          | mode <code>hc</code> , attribute <code>cmd=</code> | Extra exec channel o  |
| <code>open(s, "hs", "path=...", "w")</code>    | mode <code>hs</code> plus access chars             | SFTP file or director |
| <code>open(host, "hs", "path=...", "w")</code> | mode <code>hs</code> on a host string              | New session plus SF   |
| leading integer                                | attribute (timeout)                                | Connect timeout in    |

`channel=no` with `hc` or `cmd=` fails (error 1321). `cmd=` without `c` also fails.

## 6.2 6.2 Interactive shell

With no `cmd=`, `open(..., "h")` requests a remote PTY and shell. `net::ssh_interactive(s)` attaches the local tty (raw mode, `select()` with `&input`, CRLF for display):

```
import net

procedure main()
  s := open("user@host", "h", "key=/home/user/.ssh/id_ed25519") |
    stop(&errortext)
  ssh_interactive(s)
  close(s)
end
```

`term=`, `cols=`, and `rows=` override the PTY. Mode `"h-"` skips `known_hosts` (useful on a first connect before the key is installed). Password auth uses `password=` instead of, or in addition to, `key=`.

## 6.3 6.3 Remote command

Mode `"hc"` with `cmd=` runs one command and returns its stdout through ordinary `read()` / `reads()`. `Attrib(c, "exitstatus")` is the remote exit code once it has arrived:

```
procedure main()
  s := open("user@host", "hc", "key=/home/user/.ssh/id_ed25519",
    "cmd=uname -a") | stop(&errortext)
  while write(read(s))
  write("exit status: ", Attrib(s, "exitstatus"))
```

```
    close(s)
end
```

The host string may include a port (`user@host:2222`) or an IPv6 address (`user@[2001:db8::1]:22`). A leading integer is a connect timeout in milliseconds:

```
s := open("user@host", "hc", 5000, "cmd=true") | stop(&errortext)
```

## 6.4 6.4 Ordered stdout and stderr

Stream `reads()` see stdout only. When stdout and stderr must stay in arrival order, or when stderr should go to `&errout`, use `receive()`:

```
s := open("user@host", "hc", "key=/home/user/.ssh/id_ed25519",
          "cmd=echo out; echo err 1>&2; exit 3") | stop(&errortext)
while m := receive(s) do
  case m.addr of {
    "stdout" : writes(&output, m.msg)
    "stderr" : writes(&errout, m.msg)
    "exit"   : write(&errout, "[exit ", trim(m.msg), "]")
  }
close(s)
```

`Attrib(c, "stderr")` is the other stderr path: it drains bytes accumulated since the previous call, without preserving order against stdout. See section 8.

## 6.5 6.5 Several channels on one session

`channel=no` opens an authenticated transport. Later `open(s, "hc") / open(s, "hs")` calls `add exec`, `shell`, or `SFTP` handles on that session. Closing `s` invalidates every derived handle:

```
s := open("user@host", "h", "key=/home/user/.ssh/id_ed25519",
          "channel=no") | stop(&errortext)
c1 := open(s, "hc", "cmd=uname -a") | stop(&errortext)
c2 := open(s, "hc", "cmd=uptime") | stop(&errortext)
```

```
while write(read(c1))
while write(read(c2))
close(c1)
close(c2)
close(s)
```

Opening on an existing channel opens a sibling on the session owner, not a child of that channel.

## 6.6 6.6 SFTP

The same session carries SFTP. Mode "hs" plus `path=` opens a remote file or, if the path is a directory, lists it. `stat`, `remove`, and `rename` take the session as a leading argument:

```
s := open("user@host", "h", "key=/home/user/.ssh/id_ed25519",
          "channel=no") | stop(&errortext)

c := open(s, "hs", "path=/tmp/out.bin", "w") | stop(&errortext)
writes(c, ReadBytes("local.bin"))
close(c)

info := stat(s, "/tmp/out.bin") | stop(&errortext)
write("remote size: ", info.size)

d := open(s, "hs", "path=/tmp") | stop(&errortext)
while write(read(d))
close(d)

rename(s, "/tmp/out.bin", "/tmp/out.bin.bak")
remove(s, "/tmp/out.bin.bak")
close(s)
```

## 6.7 6.7 Concurrent hosts

Each connection is blocking. Concurrent hosts use Unicon `thread`, one thread per in-flight session:

```

procedure run_cmd(host, key)
  local s, line
  s := open(host, "hc", "key=" || key, "cmd=uname -a") | {
    write(&errout, host, ": ", &errortext)
    fail
  }
  while line := read(s) do write(host, ": ", line)
  close(s)
end

procedure main()
  local host, key
  key := "/home/user/.ssh/id_ed25519"
  every host := "user@alpha" | "user@beta" | "user@gamma" do
    thread run_cmd(host, key)
  end
end

```

Channels that share a session serialize on the session mutex ([section 10.1](#)). Separate sessions do not.

## 7. Error handling

Connection and channel failures **fail**, they do not **runerr**. This matches the SSL connect path: set **&errortext**, tear down what was allocated, and fail, so callers write **if s := open(...)** then ... else .... Runtime errors are reserved for type mistakes (passing a non-SSH file to **open(s, "hc")**, a bad **Attrib** name on an SSH file).

Error numbers, defined in **src/runtime/data.r**:

| Number | Text                            | Typical cause                                                          |
|--------|---------------------------------|------------------------------------------------------------------------|
| 1320   | SSH error                       | Connect failure, allocation, bad host string                           |
| 1321   | bad ssh attribute               | Unknown name, empty value, <b>cmd=</b> without <b>c</b> , <b>chann</b> |
| 1322   | SSH authentication error        | No method succeeded                                                    |
| 1323   | SSH host key verification error | Server not in <b>known_hosts</b> (when verifying)                      |
| 1324   | SSH channel error               | Channel open, I/O, or a closed/cascaded handle                         |
| 1325   | SFTP error                      | SFTP subsystem, path, or transfer failure                              |

`set_ssh_errortext()` appends libssh's own message where available. Blocking libssh calls run under `DEC_NARTHREADS`; `&errortext` is set only after the thread is re-registered, because Unicon allocation is not allowed while unregistered.

## 8 8. I/O: streams, `Attrib()`, and `receive()`

Three tiers, layered simple-default to full-fidelity. All three share one per-channel event queue populated by libssh callbacks ([section 8.4](#)).

### 8.1 8.1 Plain stream I/O

`read()` / `write()` / `reads()` / `writes()` work unmodified via the existing `u_read()` / `u_write()` dispatch, extended with an `Fs_SSH` branch. Stream reads consume **stdout only**. They are enough when the caller does not need to distinguish `stderr` or preserve cross-stream order.

Writes to a channel call `ssh_channel_write()` and retry short writes. Writes to a transport-only session (no channel) fail with

1. SFTP regular files use `sftp_write()` on the same path.

### 8.2 8.2 `Attrib()` for exit status, `stderr`, and EOF

`Attrib()` already exists generically – documented as "read/write attributes (threads, sockets, ...)" – and is already used for socket options. SSH channels add three query-only names:

- `Attrib(c, "exitstatus")` – the remote command's exit status, as an integer. **Fails until the exit-status callback has fired.** The implementation pumps the channel until that happens or the channel is done with no status. Preferring callback state over libssh's blocking `get_exit_*` helpers avoids version-sensitive fallbacks once callbacks are registered. This reuses the ordinary Unicon "fail = not ready" idiom, the same as `read()` failing at EOF.
- `Attrib(c, "stderr")` – drains **only new bytes since the last call** (a live incremental accessor, not a cumulative snapshot). Not gated on EOF: reading partial `stderr` mid-run is safe and useful. Returns a possibly empty string.

- `Attrib(c, "eof")` – 1 once the remote sent EOF, else 0. Useful when a caller is multiplexing with `select()` and needs to distinguish "no stdout yet" from "channel finished."

Unknown names on an SSH file raise 1321.

```
c := open(s, "hc", "cmd=apply-config /tmp/newconfig.json")
while line := reads(c) do process(line)
if      Attrib(c, "exitstatus") = 0 then write("ok")
else write("failed: " || Attrib(c, "stderr"))
```

### 8.3 8.3 receive() for ordered events

A command can produce interleaved stdout and stderr where the caller needs to know *when* an stderr message occurred relative to stdout. Neither `reads()` (stdout only) nor `Attrib(c, "stderr")` (a separate drain) can preserve that.

`receive(c)` on an SSH channel yields the next queued event as a `posix_message` whose `addr` is "stdout", "stderr", or "exit" and whose `msg` is the payload. The exit status is kept as a **string** so `receive()`'s return type stays uniform across every kind of file it can be called on; Unicon's = already coerces numeric strings.

```
c := open(s, "hc", "cmd=process_records")
while m := receive(c) do
  case m.addr of {
    "stdout" : write("OUT: " || m.msg)
    "stderr" : write("ERR: " || m.msg)
    "exit"   : write("done, status " || m.msg)
  }
```

`receive()` fails once the channel is exhausted.

### 8.4 8.4 Arrival-order queue

Preserving cross-stream order requires libssh's **callback API** (`ssh_channel_callbacks`, with a `channel_data_function` receiving an `is_stderr` flag) – not `ssh_channel_poll()` / `ssh_channel_read()`, which buffer stdout and stderr separately and lose interleaving. Callbacks are registered **before** the channel is opened so early data cannot land in libssh's own buffers.

Each callback appends a tagged chunk (`SSH_CHUNK_STDOUT / STDERR / EXIT`) to a heap-allocated linked list on the `SSHfile`. Callbacks can fire inside blocking libssh calls made while the thread is unregistered (`DEC_NARTHREADS`), where Unicorn allocation is not allowed. `ssh_pump()` drives `ssh_channel_poll_timeout()` so the callbacks run. Stream reads consume stdout chunks; `Attrib(c, "stderr")` drains stderr chunks; `receive()` pops whatever is at the head. One mechanism, three accessors.

## 9 9. Interactive shells and `select()`

### 9.1 9.1 Remote PTY

An interactive channel requests a PTY before `shell`. Size and `TERM` are described in [section 4.3](#). The session file is also marked `Fs_Socket` so existing socket dispatch (including `select()` and `get_fd()`) applies.

### 9.2 9.2 Partial-line reads()

Interactive prompts have no trailing newline. The original socket line helper (`sock_getstrg`) waits for `\n` and hung on a banner or prompt. SSH therefore has its own `ssh_getstrg()`: it reads stdout bytes from the queue and returns a partial line at EOF or when the caller's buffer fills, without blocking for a newline that will never come. A seen-but- unconsumed newline is remembered in `nl_pending` (libssh has no `MSG_PEEK`). `reads(s, n)` on a channel is a byte read from the same queue and is the right primitive for an interactive loop.

### 9.3 9.3 `select()` readiness

`select()` on an SSH file must not hang when libssh already buffered the banner during `open()`. `ssh_file_pending()` nonblocking-pumps the session and reports ready when stdout (or EOF) is queued. Only stdout matters here: stream `reads()` do not consume stderr/exit chunks, so a nonempty queue of those alone must not make `select()` claim the file is readable. `get_fd()` returns `ssh_get_fd(session)` so the kernel fd participates in the same `select()` set as ordinary sockets.

On Windows, `select(&input)` and `Attrib(f, "tty=raw") / "tty=sane"` were added so an interactive client can mux the console with the channel on both Unix and Windows console `iconx`. Those `tty` attributes are general (they apply to `&input`), not SSH-specific.

## 9.4 9.4 `net::ssh_interactive()` and `ussh`

`uni/lib/ssh.icn` provides `ssh_interactive(s)`: put the local `tty` in raw mode, mux `&input` with the remote channel via `select()`, map lone LF to CRLF when copying remote output to a raw local terminal (needed because raw mode disables `ONLCR`), and restore the `tty` before returning. It does not close `s`. The `ussh` demo (`uni/progs/ussh.icn`) is a small OpenSSH-like client built with the other `uni/progs` demos: interactive shell, or a one-shot remote command via `receive()`.

```
s := open("user@host", "h", "key=...") | stop(&errortext)
ssh_interactive(s)
close(s)
```

## 10 10. Concurrency and channel lifecycle

### 10.1 10.1 Shared mutex

Every `b_file` already has a `mutexid`, and all I/O in `fsys.r` already locks around it. When a channel is created via `open(s, "hc") / open(s, "hs")`, it does **not** allocate a fresh mutex id – it copies the session's existing `mutexid`. Every channel sharing a session then serializes through the same lock via the existing locking calls in `u_read / u_write / receive / Attrib`. `libssh` sessions are not safe for uncoordinated concurrent access; two threads each holding a different channel on the same session block each other during actual I/O because they contend for the same mutex.

### 10.2 10.2 Child tracking and cascade close

A Unicon list of `b_file` pointers was considered so the moving collector would relocate them. `SSHfile` is `malloc'd` and therefore does not move, so the session owner holds a C linked list of child `SSHfile * (children / next / parent)`. No Unicon list is required.



`close(s)` on the session walks that list, force-closes every remaining channel and SFTP handle (same close-hook logic as an explicit `close()`), marks each child `closed`, clears its queue, and then disconnects and frees the session. The child's Unicon `b_file` stays alive until its own `close()`, but is unusable immediately – no dangling libssh objects, no readable leftovers. Closing a channel unlinks it from the owner and frees only that channel.

The SFTP subsystem is created lazily, once, on the session owner (`ssh_owner_sftp()`) and shared by every SFTP file or directory on that session. It is freed when the session closes.

## 10.3 10.3 Blocking model

I/O uses blocking libssh calls plus Unicon's existing `thread` mechanism – one thread per in-flight host or channel. Chosen over non-blocking libssh plus `select()` as the *primary* model: it requires no partial-read/retry state machines. `select()` readiness ([section 9.3](#)) is implemented so interactive muxing works; it is not a full non-blocking I/O API.

## 11 11. SFTP

Three shapes, each mapped onto an existing Unicon pattern.

### 11.1 11.1 File transfer – `read()` / `write()`

```
c := open(s, "hs", "path=/tmp/firmware.bin", "w")
while writes(c, ReadBytes(local_fw))
close(c)
```

Mode `"hs"` is SFTP; read/write/append intent comes from the usual mode characters (`r` / `w` / `a` / `b`), either in the same string (`"hsw"`) or as a trailing mode-only token. `path=` is required. Backed by `sftp_open()` / `sftp_read()` / `sftp_write()` / `sftp_close()`. Structurally a plain file – another variant feeding the existing dispatch. SFTP regular files are **not** marked `Fs_Socket`: they are byte streams, not select-able channels.

## 11.2 11.2 Directory listing

Local `open()` on a path that turns out to be a directory transparently switches to `opendir()` / `readdir()`, sets `Fs_Directory`, and `reads()` yields one entry name per call. SFTP does the same: `open(s, "hs", "path=/tmp/configs")` checks the remote path with one `sftp_stat()`; if it is a directory, `sftp_opendir()` / `sftp_readdir()` provide the same `Fs_Directory` behavior.

## 11.3 11.3 Metadata – `stat()`, `remove()`, `rename()`

These extend existing builtins via the leading-optional-file argument idiom, rather than introducing `sftp_remove()` names. The signature change is backward-compatible: a string first argument is still the local-filesystem operation.

- `remove(s, path)` – `sftp_unlink` when `s` is an SSH session or channel file.
- `rename(s, from, to)` – `sftp_rename`. `rename` gained a third argument for this form; `rename(s1, s2)` is unchanged.
- `stat(c)` – `sftp_fstat` on an already-open SFTP file.
- `stat(s, path)` – a single `sftp_stat` request, without paying for `sftp_open()` + `sftp_fstat()`. Both entry points are worth supporting rather than just one.

```
if info := stat(s, "/tmp/firmware.bin") then
  if info.size = expected_size then pull_it()
```

`sftp2rec()` populates the same `posix_stat` record as `stat2rec()`. SFTP's attribute set is protocol-version dependent; fields the server did not send are left **null** rather than reported as zero, so callers can tell "absent" from "genuinely zero." SFTP carries `atime/mtime`, not `ctime`. Size, `uid/gid` (name or numeric), permissions-as-mode-string, and file type (`d` / `l` / `c`) are filled when the corresponding flags are present.

## 12 12. Build integration

`configure -disable-ssh` turns the feature off. Otherwise `CHECK_LIBSSH` in `aclocal.m4` probes for `<libssh/libssh.h>` and `ssh_new`, setting `HAVE_LIBSSH`. The probe looks under `/usr/local` (FreeBSD) and `/opt/homebrew` (macOS) when those prefixes contain the headers, so later compiles do not miss `libssh/libssh.h`. `-with-libssh=DIR` overrides the prefix. A required `-enable-ssh` without the library is a configure error; optional (default) absence degrades gracefully, as with other optional libraries.

`&features` reports `secure shell` when the library was found. `tests/posix/Makefile` skips `ssh` when the feature is absent. CI installs `libssh-dev` (or the platform equivalent) so the support builds and the deterministic half of the test runs. README install notes list `libssh-dev` / `libssh-devel` / `brew install libssh` alongside the other optional libraries.

Guards are `#if HAVE_LIBSSH` throughout `rstructs.h`, `rmacros.h`, `feature.h`, `sys.h`, `fsys.r`, `rposix.r`, `fxposix.ri`, and `fmisc.r`.

## 13 13. Status and remaining work

**Keyboard-interactive authentication** and an explicit agent attribute. The `publickey_auto` fallback already talks to a running agent.

**Non-blocking I/O as the primary concurrency model.** `select()` readiness is implemented; the I/O calls themselves still block. A `ssh_set_blocking(sess, 0)` plus resume state machine is the obvious next step if thread-count pressure appears.

**SSH server.** `libssh` can do it; nothing in the runtime exposes it. The mode character and file-status bit leave room, but the accept/listen path is not designed.

**Default channel type.** Shell-by-default shipped for mode "h". Scripted use uses "hc" with `cmd=`. Reconsidering the default remains open and would be a compatibility break.

**Timeouts.** TCP-connect timeout is the leading integer attribute. There is no separate handshake+auth timeout beyond what that covers, and no exec timeout – a stuck command blocks until `close()`, matching ordinary blocking sockets.

**Trust-on-first-use.** Verification is check-only (`ssh_session_is_known_server`). A new host fails with 1323 unless the caller uses `h-` or installs the key in

`known_hosts` out of band. The runtime does not write the store.

`~ in key=`. Not expanded. The helper and `ussh` expand `~/` themselves.

## 14 Appendix: test coverage

`tests/posix/ssh.icn` is deterministic without a server: it checks that `&features` reports `secure shell`, and that bad attributes, `channel=maybe`, `channel=no` combined with `hc/cmd=`, `cmd=` without mode `c`, and a connect to a closed local port all fail with `&errortext` set. A live round-trip runs only when `UNICON_SSH_TESTHOST` is set (host or `user@host`), with optional `UNICON_SSH_TESTPORT`, `UNICON_SSH_TESTKEY`, `UNICON_SSH_KNOWNHOSTS`, and `UNICON_SSH_SFTPPATH`. The live path covers `exec stdout`, `Attrib(c, "exitstatus")`, `receive()` event tags, partial-line `reads()` of a `printf` with no newline, and SFTP `write/read/stat/remove` of a binary payload.

`tests/posix/Makefile` skips the test when the feature is absent. Expected default output is `tests/posix/stand/ssh.std`.

## 15 References

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